

Does the Risk Model Still Matter Once the Investor's Rules Are In?

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Blend-Ed

Background

Anyone investing money faces one decision: how much to put in each asset. Portfolio optimisation answers it by estimating each asset's expected return and risk from past prices and solving for the best mix. In practice the estimates are noisy, and real investors add rules the textbook ignores: a limit on the number of holdings, a cash floor, sector caps, trading costs.

Real-world investing imposes constraints on portfolio optimisation. We ask how the choice of risk model changes the outcome once those constraints are fixed, and whether any model earns its complexity over the simplest alternative, equal weight.

What we ran. 5 investor rule-sets $\times$ 5 risk models (Markowitz, Ledoit–Wolf, robust, CVaR, equal weight), walk-forward over 2016–2025 with trading costs.

Limitations. One decade, one market (US large caps), no short selling or leverage, and a universe chosen with hindsight: all 40 stocks still trade today.

1. Experimental Setup

- **Universe:** 43 assets: 40 US large caps (four per GICS sector), a bond ETF, gold, cash.
- **Prices:** daily, 2008–2025. Tune on 2011–2015, test on 2016–2025.
- **Rule-sets:** five archetypes, values derived from Vanguard allocations and the 2022 Survey of Consumer Finances.

Archetype	CVaR limit start→end	max holdings	$w_{\max}$	cash floor	sector caps
Preserver	1.0→0.5%	20	8%	10%	IT 20%, Energy 10%
Steady Hand	1.5→0.7%	15	10%	5%	–
Builder	2.0→1.0%	12	12%	2%	–
Maverick	2.8→1.4%	8	20%	0%	–
Ethical	1.5→0.7%	15	10%	5%	excl. tobacco, oil, defence

2. The Portfolio Problem

$\mathbf{w} \in \mathbb{R}^n$
 $z_a \in \{0, 1\}$
 $b_a, s_a \geq 0$
 $\hat{\mu}, \hat{\Sigma}, r_s$
 $h = 21, \tau = 10 \text{ bp}$
 $\rho_{\max}(t)$

weights: fraction of the portfolio in each of n assets
whether asset a is held
amount bought and sold, relative to the current weights $\mathbf{w}^{\text{prev}}$
expected daily return, covariance, and day- s return vector, estimated from the last $|S|$ days
expected holding period (days); cost per unit traded
risk limit, tightening linearly as the horizon shrinks

$$\max_{\mathbf{w}, z, b, s} h \hat{\mu}^\top \mathbf{w} - \tau \sum_a (b_a + s_a) \quad \text{s.t.} \quad \rho(\mathbf{w}) \leq \rho_{\max}(t)$$

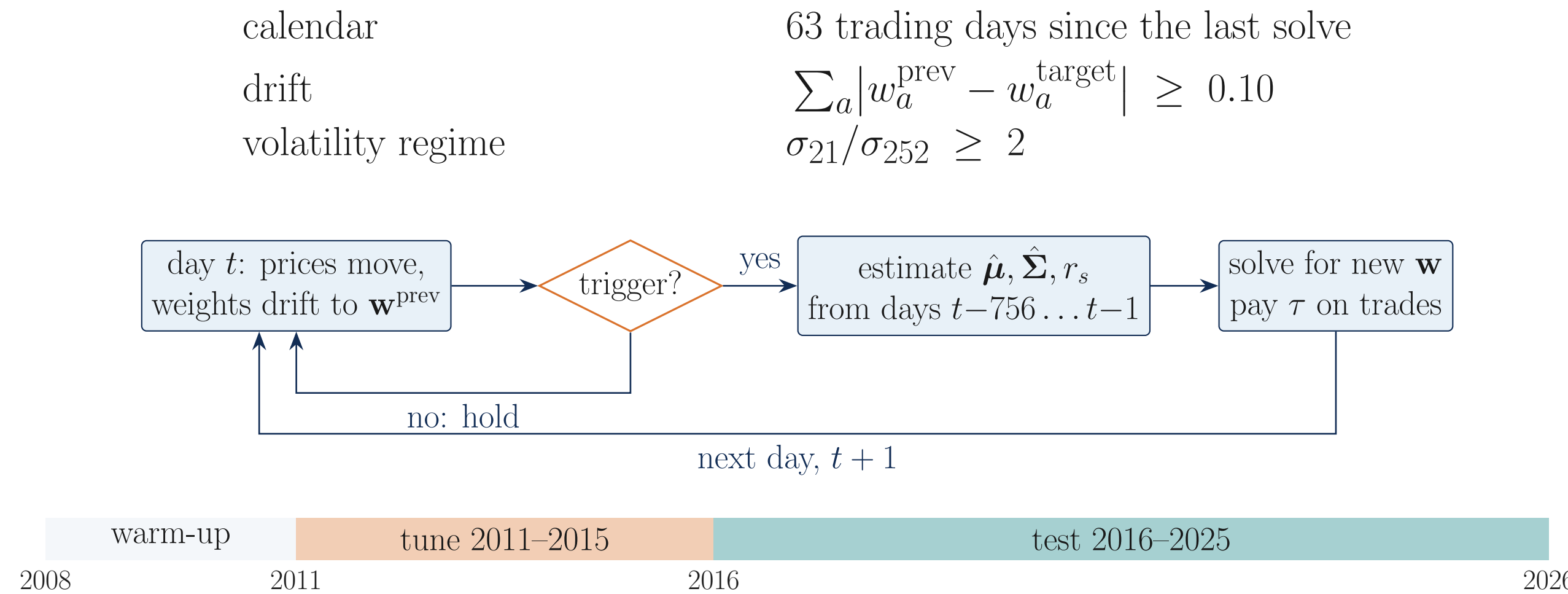
$\sum_a w_a = 1$ budget
 $w_a = w_a^{\text{prev}} + b_a - s_a$ trade balance
 $w_a^{\min} z_a \leq w_a \leq w_a^{\max} z_a$ minimum and maximum position
 $\sum_a \notin \text{cash } z_a \leq K$ cardinality: at most K names
 $\sum_{a \in \text{sector } k} w_a \leq c_k$ sector caps
 $w_{\text{cash}} \geq c_{\min}$ cash floor

3. Five Risk Models

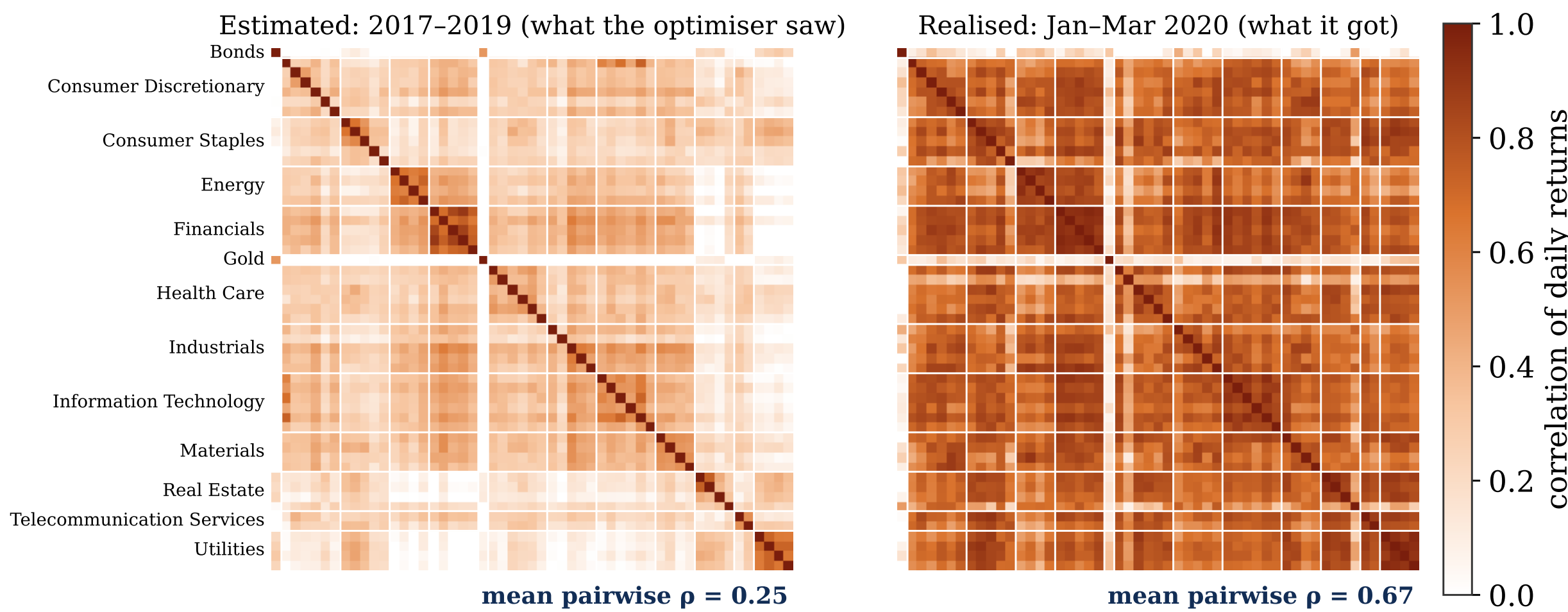
Markowitz $\rho(\mathbf{w}) = \mathbf{w}^\top \hat{\Sigma} \mathbf{w}$
Markowitz + Ledoit–Wolf $\rho(\mathbf{w}) = \mathbf{w}^\top \hat{\Sigma}_{\text{LW}} \mathbf{w}, \quad \hat{\Sigma}_{\text{LW}} = (1 - \delta) \hat{\Sigma} + \delta \bar{\sigma}^2 I$
Robust $\rho(\mathbf{w}) = \mathbf{w}^\top \hat{\Sigma} \mathbf{w}, \quad \hat{\mu}^\top \mathbf{w} \rightarrow \hat{\mu}^\top \mathbf{w} - \kappa \sqrt{\mathbf{w}^\top \Omega \mathbf{w}}, \quad \Omega = \hat{\Sigma}/|S|$
CVaR $\rho(\mathbf{w}) = \zeta + \frac{1}{(1 - \alpha)|S|} \sum_{s \in S} u_s, \quad \alpha = 0.95$
 $u_s \geq -r_s^\top \mathbf{w} - \zeta, \quad u_s \geq 0$
Equal weight $w_a = 1/N$

4. When the Portfolio Is Reweighted

Starting from the weights currently held, $\mathbf{w}^{\text{prev}}$, the portfolio is re-solved when any of the following holds ($\mathbf{w}^{\text{target}}$ = weights chosen at the last solve; $\sigma_{21}, \sigma_{252}$ = market volatility over the last 21 and 252 days):

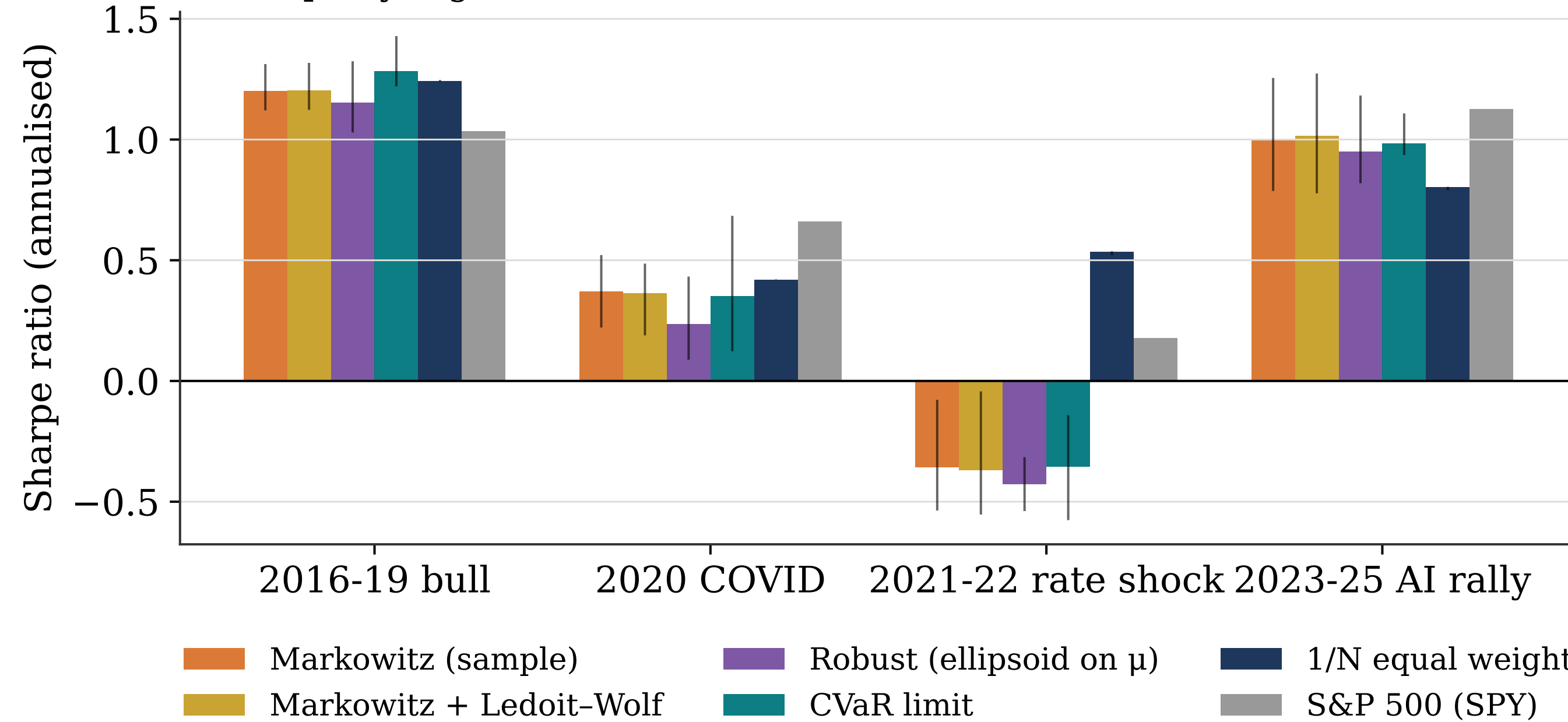


5. Results I: When the Inputs Fail



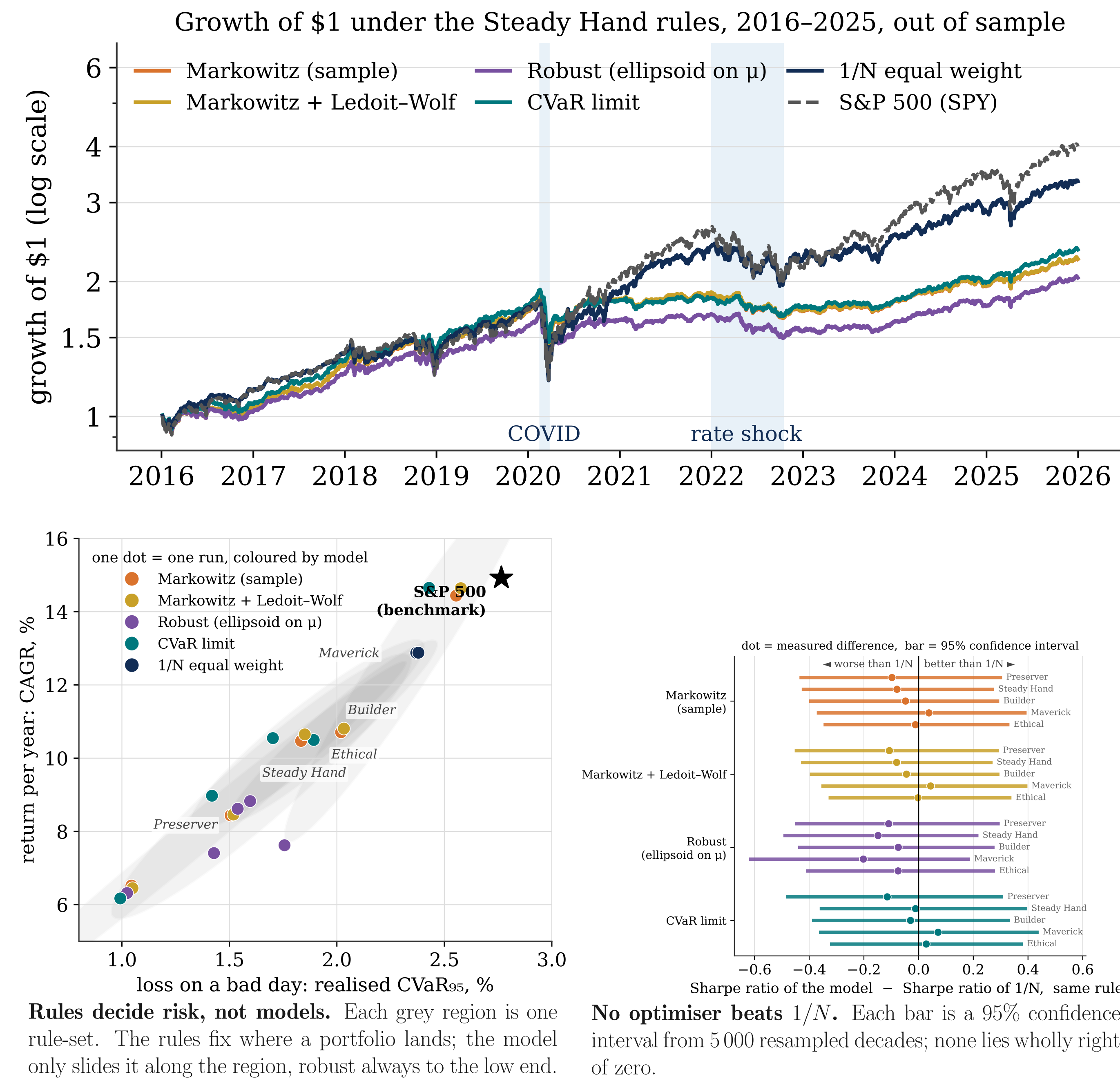
One event turned every pairwise estimate wrong at once: mean correlation 0.25 $\rightarrow$ 0.67. Every model in box 3 was fed the left-hand matrix and traded in the right-hand one.

Sharpe by regime. Bar = mean over the five rule-sets, whisker = min-max



2020: all models fell together, as the matrices predict. 2021–22: every optimiser negative, 1/N positive. A three-year lookback had loaded the optimisers onto the 2019–21 winners just before rates rose.

6. Results II: Rules vs Models



Rules decide risk, not models. Each grey region is one rule-set. The rules fix where a portfolio lands; the model only slides it along the region, robust always to the low end.

No optimiser beats 1/N. Each bar is a 95% confidence interval from 5000 resampled decades; none lies wholly right of zero.

Model	CAGR %	Sharpe [95% CI]	Max DD %	CVaR ₉₅ %
Markowitz (sample)	10.1	0.67 [0.14, 1.35]	-25	1.79
Markowitz + Ledoit–Wolf	10.2	0.67 [0.14, 1.35]	-25	1.81
Robust (ellipsoid on μ)	7.8	0.59 [0.03, 1.29]	-22	1.47
CVaR limit	10.2	0.70 [0.17, 1.40]	-25	1.69
1/N equal weight	12.9	0.71 [0.19, 1.36]	-33	2.37
S&P 500 (SPY)	14.9	0.74 [0.20, 1.38]	-34	2.77

What the optimisers bought: 3–5 points less CAGR for ≈ 8 points less drawdown than 1/N, at the same Sharpe. They delivered the risk target they were given.

CAGR = average growth per year. Sharpe = return above cash per unit of volatility. Max DD = largest peak-to-trough fall. CVaR₉₅ = mean loss on the worst 5% of days.

7. Conclusions

- **The rules matter more than the model.** Four risk models within 0.04 Sharpe of each other and of 1/N, inside bootstrap noise.
- **CVaR is the best optimiser and the cheapest** (linear, no covariance); still not better than 1/N.
- **Robust is consistently worst:** guarding against the worst-case mean doubles up on rules that already limit concentration.
- **Nothing beats the S&P 500** in a mega-cap decade whose winners every rule-set caps. The optimisers bought drawdown, not return.

Interactive version: pick an investor, travel back to 2016 and watch each model trade.



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